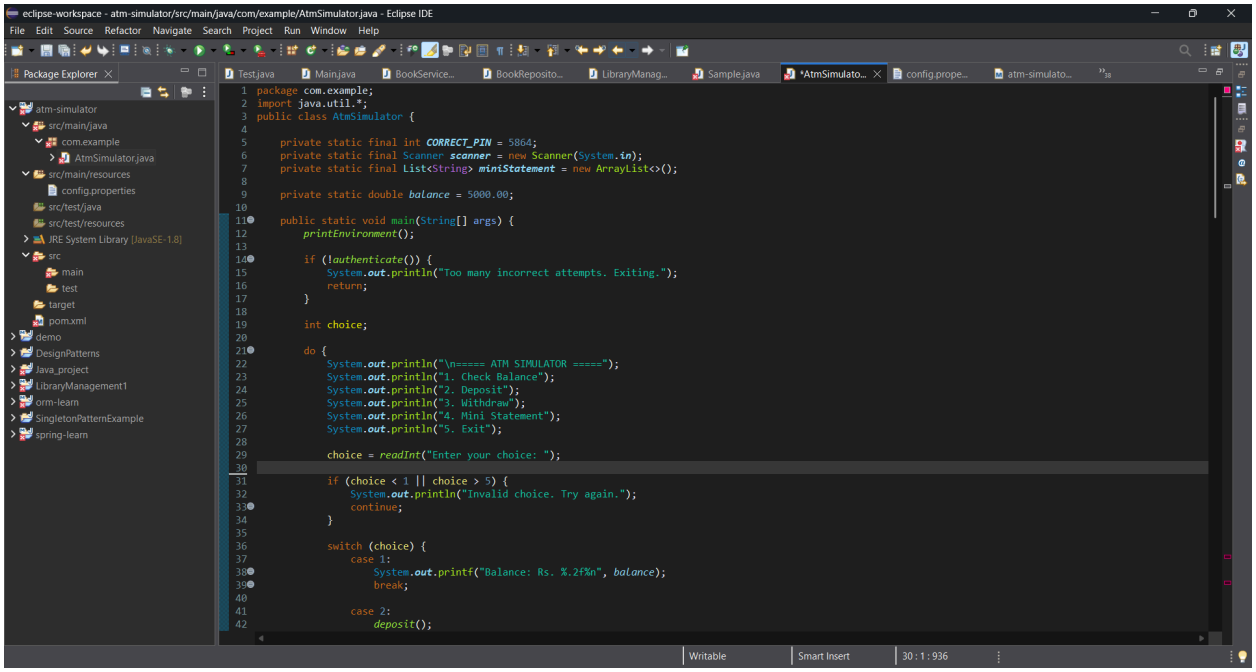



```
package com.example;
import java.util.*;
public class AtmSimulator {
    private static final int CORRECT_PIN = 1234;
    private static final Scanner scanner = new Scanner(System.in);
    private static final List<String> miniStatement = new ArrayList<>();

    private static double balance = 5000.00;

    public static void main(String[] args) {
        printEnvironment();

        if (!authenticate()) {
            System.out.println("Too many incorrect attempts. Exiting.");
            return;
        }

        int choice;

        do {
            System.out.println("\n===== ATM SIMULATOR =====");
            System.out.println("1. Check Balance");
            System.out.println("2. Deposit");
            System.out.println("3. Withdraw");
            System.out.println("4. Mini Statement");
            System.out.println("5. Exit");

            choice = readInt("Enter your choice: ");
```

```
            // Continue skips the remaining menu processing
            // when the choice is invalid.
            if (choice < 1 || choice > 5) {
                System.out.println("Invalid choice. Try again.");
                continue;
            }

            switch (choice) {
                case 1:
                    System.out.printf("Balance: Rs. %.2f%n", balance);
                    break;

                case 2:
                    deposit();
                    break;

                case 3:
                    withdraw();
                    break;

                case 4:
                    showMiniStatement();
                    break;

                case 5:
                    System.out.println("Thank you for using the ATM.");
                    break;

                default:
                    System.out.println("Invalid choice.");
            }

        } while (choice != 5);

        scanner.close();
    }

    private static boolean authenticate() {
        for (int attempt = 1; attempt <= 3; attempt++) {
            int pin = readInt("Enter your 4-digit PIN: ");

            if (pin == CORRECT_PIN) {
                System.out.println("Login successful.");
                return true;
            }

            System.out.println(
                "Incorrect PIN. Attempts remaining: " + (3 - attempt)
```

```
            );
        }

        // Exit authentication after three failed attempts.
        return false;
    }

    private static void deposit() {
        double amount = readAmount("Enter deposit amount: ");

        if (amount <= 0) {
            System.out.println("Amount must be greater than zero.");
            return;
        }

        balance += amount;
        miniStatement.add(String.format("Deposit: +Rs. %.2f", amount));

        System.out.println("Deposit successful.");
    }

    private static void withdraw() {
        double amount = readAmount("Enter withdrawal amount: ");

        if (amount <= 0) {
            System.out.println("Amount must be greater than zero.");
            return;
        }

        if (amount > balance) {
            System.out.println("Insufficient balance.");
            return;
        }

        balance -= amount;
        miniStatement.add(String.format("Withdrawal: -Rs. %.2f", amount));

        System.out.println("Please collect your cash.");
    }

    private static void showMiniStatement() {
        System.out.println("\n===== MINI STATEMENT =====");

        if (miniStatement.isEmpty()) {
            System.out.println("No transactions yet.");
            return;
        }

        // Enhanced-for loop
```

```
    for (String transaction : miniStatement) {
        System.out.println(transaction);
    }

    System.out.printf("Current Balance: Rs. %.2f%n", balance);
}

private static int readInt(String message) {
    while (true) {
        System.out.print(message);
        String input = scanner.nextLine().trim();

        try {
            return Integer.parseInt(input);
        } catch (NumberFormatException e) {
            System.out.println("Please enter a valid whole number.");
        }
    }
}

private static double readAmount(String message) {
    while (true) {
        System.out.print(message);
        String input = scanner.nextLine().trim();

        try {
            return Double.parseDouble(input);
        } catch (NumberFormatException e) {
            System.out.println("Please enter a valid numeric amount.");
        }
    }
}

private static void printEnvironment() {
    try (var input = AtmSimulator.class
            .getClassLoader()
            .getResourceAsStream("config.properties")) {

        if (input == null) {
            System.out.println("Environment: not configured");
            return;
        }

        var properties = new java.util.Properties();
        properties.load(input);

        System.out.println(
            "Environment: " + properties.getProperty("app.environment")
        );
```

```
        } catch (java.io.IOException e) {
            System.out.println("Could not load environment configuration.");
        }
    }
}
```

POM.XML file -->

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         https://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>atm-simulator</artifactId>
    <version>1.0-SNAPSHOT</version>
    <packaging>jar</packaging>

    <properties>
        <maven.compiler.release>21</maven.compiler.release>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
        <app.environment>dev</app.environment>
    </properties>

    <build>
        <resources>
            <resource>
                <directory>src/main/resources</directory>
                <filtering>true</filtering>
            </resource>
        </resources>

        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.13.0</version>
            </plugin>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.4.2</version>
                <configuration>
```

```
            <archive>
              <manifest>
                <mainClass>com.example.AtmSimulator</mainClass>
              </manifest>
            </archive>
          </configuration>
        </plugin>
      </plugins>
    </build>

    <profiles>
      <profile>
        <id>dev</id>
        <activation>
          <activeByDefault>true</activeByDefault>
        </activation>
        <properties>
          <app.environment>dev</app.environment>
        </properties>
      </profile>

      <profile>
        <id>prod</id>
        <properties>
          <app.environment>prod</app.environment>
        </properties>
      </profile>
    </profiles>

</project>
```